

Performance Evaluation of Computer Systems and Networks

Simulation

MSc in Computer Engineering
MSc in Artificial Intelligence and Data Engineering

Academic Year 2026/2027

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UNIVERSITÀ DI PISA

Who am I?

Dr. Ing. Giovanni Nardini – Assistant Professor



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- Office hours
 - Mondays, 10-12
 - Please email me to schedule a meeting
 - Either in presence or online

Course organization

Teaching module on simulation and practical aspects of performance evaluation

- Lectures (approx. 18-20 hours)
- Labs (approx. 10-12 hours)

Please bring your own laptop!

- Text editor, C++ compiler
- OMNeT++ simulation framework ([download](#), [install guide](#))
 - You may use [opp env](#) for quicker installation
- Spreadsheet software, such as Excel or Calc

Teaching material

- Slides with notes, made available via MS Teams
- Recordings from a.y. 2025/2026

Bibliography

- A.M. Law, W.D. Kelton, "Simulation modeling and analysis", McGraw-Hill
- R. Jain, "The art of computer systems performance analysis", Wiley
- S.M. Ross, "Introduction to probability and statistics for engineers and scientists", Elsevier
- OMNeT++ simulation manual ([link](#))

Bring your laptop, equipped with:

- one **working C++ compiler** (whichever you prefer, I will use g++ - you can use it by downloading the MinGW environment in Windows). No specific IDE is required, indeed you can do exercises without having an IDE at all (a text editor is fine).
- At some point, you will need the **OMNeT++ framework**, so you can already try to download and install it.
- A **spreadsheet** tool to analyze data, and apply some of the concepts learnt in the lectures by Prof. Stea.

The part on analysis of output data can be found on both Law-Kelton's and Ross's books.

Contents

- Introduction to simulation
 - Why should we use simulation?
 - Types of simulators
 - How to build a (good) simulator
 - Components of a (discrete-event) simulator
 - Random number generation
 - How to do a simulation analysis
 - Simulation workflow
 - Working with the real thing
 - The OMNeT++ framework
 - How to gather meaningful data
 - Steady-state analysis
 - Statistically-sound output results
- Building blocks of an OMNeT++ model
 - How to record statistics
 - Advanced concepts (e.g., modularity, RNGs)
 - Debugging
- Group project**
Groups composed of 3 (three) students
Modeling, implementation and analysis

Introduction to the course

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We'll talk about simulation because this is a relevant method to **practically** do performance evaluation of a real system.

- How to build a piece of software which we call simulator, which has peculiar characteristics with respect to classic software, such as RNG, for example.
- How to run simulations **properly**, specifying the correct workflow you should follow in order to do performance evaluation through simulation
- How to extract meaningful results from the simulation
- In between, we will learn how to use OMNeT++: one of the most popular simulation frameworks in both academia and industry. It is based on C++, so you should brush up on your knowledge about C++.
- How to create a simulation model, record statistics, and also will give you some hints on how to debug your simulation model

The **group project** will build up on the above contents, and will consist of modeling a given system, implementing it within OMNeT++ and then perform the analysis of the system using that implemented model